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### **Statement of Interest: Computer Vision Methods for Ecology**

#### **A. Research Problem**

Gully erosion is among the most severe and fastest-moving forms of land degradation in southeastern Nigeria. In Anambra, Imo, and Enugu states, active gullies now consume agricultural land and infrastructure at rates that challenge manual monitoring entirely. The Nigerian Erosion and Watershed Management Project has mapped over 3,000 erosion-prone sites, yet systematic real-time tracking at the individual gully level remains largely absent. Field surveys are slow, expensive, and dangerous, and the degradation outpaces assessment.

The research problem I want to address is: can computer vision methods applied to multi-temporal satellite and drone imagery automate the detection, classification, and volumetric change estimation of active gullies in southeastern Nigeria at a resolution and frequency that makes real-time early warning systems feasible?

This matters beyond Nigeria. Gully erosion accounts for an estimated 10 to 15 percent of global sediment yield and is accelerating under changed precipitation regimes across sub-Saharan Africa. A computer vision pipeline trained on Nigerian gully morphology could transfer to analogous landscapes in Ethiopia, Kenya, and Madagascar with relatively low retraining cost.

#### **B. Data**

My MSc thesis work in Lagos State uses Landsat 8, Sentinel-2, and SRTM DEM data processed in ArcGIS Pro and QGIS to analyse urbanisation impacts on green spaces and drainage patterns from 2000 to 2020. This work has given me direct experience with multi-temporal raster preprocessing, supervised land cover classification, and change detection workflows.

For the gully erosion project specifically, I plan to work with three data sources. First, Sentinel-2 multispectral imagery at 10m resolution, freely available via the Copernicus Open Access Hub, covering Anambra and Imo states at monthly intervals from 2017 to present. Second, Planet Labs SkySat imagery at 0.5m resolution, which I am applying for through the Planet Education and Research Programme. Third, existing gully inventory shapefiles from the Nigeria Erosion and Watershed Management Project (NEWMAP), which provide labelled ground-truth polygons for approximately 600 mapped gully sites.

By the time the summer school begins I will have the Sentinel-2 archive downloaded and preprocessed, the NEWMAP inventory cleaned and georeferenced, and an initial binary segmentation dataset of approximately 400 labelled image patches. Planet imagery access will determine whether I can add a high-resolution layer before or after the course.

### **C. Impact and Outlook**

The immediate scientific impact is a reproducible computer vision pipeline for gully detection in tropical savanna landscapes, validated against field-mapped ground truth and published as open-source code. This fills a methodological gap: most existing erosion detection literature focuses on European or arid-zone contexts, and very few studies have applied deep learning segmentation to the specific morphological signatures of Nigerian basement-complex gullies.

The policy impact is equally direct. The Nigerian government is currently implementing a World Bank-funded erosion control programme requiring regular monitoring of intervention sites. An automated satellite-based detection tool would reduce monitoring cost significantly and allow programme managers to prioritise interventions based on real-time change data rather than annual field surveys.

The educational dimension connects to my broader work. I am currently building Verdant, an ecological simulation game that teaches land restoration through gameplay grounded in published science. Computer vision methods for real-time gully monitoring would feed directly into the data layer underpinning that simulation, closing the loop between satellite observation, scientific modelling, and public ecological literacy.